VANGALAPUDI DHANYA

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Results-driven Data Engineer with 4 years of experience building data and machine learning solutions on Google Cloud, including anomaly detection with BigQuery ML and LLM prompt engineering with Gemini on Vertex AI. Google Professional Machine Learning Engineer certified, with a strong focus on data quality, metric validation and clearly communicating technical findings to stakeholders. Seeking a role evaluating and improving AI model outputs on data science and quantitative problems.

CORE SKILLS

- **Machine Learning & AI:** BigQuery ML, Anomaly Detection, Classification Modelling, Exploratory Data Analysis (EDA), Vertex AI (Prompt Management, Gemini), LLM Prompt Engineering
- **Data Quality & Analytics:** Data Reconciliation, Data-Quality Monitoring, Metric Validation, Discrepancy Root-Cause Analysis, Looker Dashboards
- **Programming & Data:** Python, SQL, BigQuery, Snowflake, Spark, Airflow, Data Modelling, Data Lake Architecture
- **Cloud & Data Platforms:** GCP (Dataproc, Composer, GCS, Cloud Functions, Data Fusion, Dataform), AWS (S3, EMR), ETL/ELT Pipelines
- **Engineering Practices:** Git, Docker, CI/CD Pipelines, New Relic, SMTP-based Alerting, Reusable Python Modules

WORK EXPERIENCE

Data Engineer
Insight Global | Bangalore (Jun 2025 – Present)

- Designed and implemented an **anomaly detection system** leveraging **BigQuery ML** and **Cloud Functions**, enabling proactive identification of irregular KPI shifts and data-quality issues.
- Partnered with business stakeholders to troubleshoot metric deviations and ensure consistent definitions across dashboards and data products.
- Improved data quality by implementing reconciliation logic between MyServicesDrill and Home Services datasets, reducing mismatch rates and improving reporting accuracy.
- Designed and optimized end-to-end ETL pipelines to process **high-volume retail and services data**, improving data freshness and reliability across enterprise dashboards.

Data Engineer
NewsCorp | Bangalore (Nov 2023 – Jun 2025)

- Automated the migration of SQL scripts from a Git repository to Dataform using **Vertex AI Prompt Management** and the **Gemini-1.5 Flash** model, reducing migration time by 70% and significantly enhancing quality and speed of migration.
- Created a Python module and optimized codebases, eliminating manual troubleshooting, reducing debugging time by 60%, and enhancing scalability across multiple Airflow DAGs for faster issue diagnosis.
- Built a custom Airflow operator to automate Dataform workflow execution, reducing DAG complexity by 50% and enhancing overall operational efficiency by 25%.
- Designed a custom Airflow operator to optimize GCS file composition, overcoming the 32-file limit by implementing an iterative merging strategy, reducing processing time by 40% and improving system robustness.
- Developed and deployed Data Fusion pipelines for seamless Salesforce and database file ingestion, consistently meeting project deadlines and improving data processing efficiency by 30%.

Data Engineer
Searce | Bangalore (Jun 2022 – Nov 2023)

- Led end-to-end development of a fully automated data migration pipeline and script transferring ~5TB of data from Snowflake to BigQuery within a tight 2-week timeline, achieving **zero data loss** and ensuring a seamless platform transition.
- Developed and implemented an automated email alerting mechanism using BigQuery and an SMTP server, enhancing monitoring and notification capabilities, which improved incident response time by 50%.
- Designed and implemented an end-to-end solution to migrate SQL Server databases to BigQuery, reducing migration costs by 30% and improving performance by 25%.
- Ideated and prepared a presentation for a Data Mesh Implementation workshop in GCP, resulting in an opportunity to present to Google Cloud leadership and receiving commendation for the insights shared, enhancing visibility within the organization.

PROJECTS

1. **Assessment and Prediction of Stressors among Healthcare Workers during COVID-19 using Machine Learning Techniques -** Collected data through offline and online surveys. Conducted Exploratory Data Analysis (EDA) and implemented five machine learning algorithms for classification and prediction of stressors among healthcare workers.
2. **Perform 'Exploratory Data Analysis' on Dataset 'Sample Superstore' -** Performed Data Cleaning, Data Visualization, and Data Analysis to identify areas for profit maximization.

EDUCATION

- **B.Tech** National Institute of Technology, Calicut **GPA:** 8.43/10 **(Jan 2018 - Jan 2022)**
- **Class XII** Sri Chaitanya Junior College **GPA:** Score: 977/1000 **(Jan 2015 - Jan 2017)**
- **Class X** Siddhartha High School **GPA:** Score: 9.5/10 **(Jan 2015 - Jan 2015)**

CERTIFICATIONS

1. Google Professional Data Engineering Certification Dec 2022
2. Google Professional Machine Learning Engineer Certification July 2024
3. Google Professional Cloud Database Engineer Certification Apr 2023

AWARDS

1. **4th Position, All India Inter-NIT Kho-Kho Tournament –** Demonstrated dedication, resilience, and teamwork in a highly competitive national event.
2. **Most Entrepreneurial Team, Climate Data Hackathon –** Recognized for innovative solutions in 'Climate Resilient Agriculture,' showcasing creativity and problem-solving skills.
3. **Employee Spotlight Award (FY'24) –** Honored for outstanding performance, teamwork, and contributions to organizational success.
4. **Star Performer of the Month (April 2024) –** Awarded for exceeding targets and displaying exceptional commitment to project success.